Bubble Sort
Selection Sort
Insertion Sort
⋮
} $O(n^2)$

Merge Sort
↳ Quick Sort
} $O(n \log n)$

Stable Sorting Algos

stable
[3a, 5, 2c, 7, 3b, 6, 2d]
[2c, 2d, 3a, 3b, 5, 6, 7]

[2c, 2d, 3b, 3a, 5, 6, 7]   Unstable

↳ $(Alice, \underline{B})(Carol, \underline{A})(Dave, \underline{A})(Eric, \underline{B})(Ken, \underline{A})$

Unstable
~~So we might have to sort again to obtain the list of~~ like this:

$(Carol, \underline{A})(Dave, \underline{A})(Ken, \underline{A})(Eric, \underline{B})(Alice, \underline{B})$

Stable.

(Carol, A) (Dave, A) (Ken, A) (Alice, B) (Eric, B)

Merge-sort is a stable sorting algo

Merge Sort.   7/2 = 3.

0 1 2 3 | 4 5 6 7
[3, 5, 7, 2, 1, 6, 3, 8]

[3, 5, 7, 2]   [1, 6, 3, 8]
[2, 3, 5, 7]   [1, 3, 6, 8]

[1, 2, 3, 3, 5, 6, 7, 8]

low = 0   high = 7
mid = $\frac{0+7}{2}$ = 3

[low, mid]   [mid+1, high]

[1, 2, 3, 4]   [3, 5, 6, 7]

[2, 3, 5, 7, 1, 3, 6, 8]
0 1 2 3 4 5 6 7

temp: [1, 2, 3, 3, 5, 6, 7, 8]

0   low   high   n-1
↳ [...... 3, 5, 7, 2, 1, 6, 3, 8 ....]

temp[0] → arr[low]
temp[1] → arr[low+1]
⋮
temp[i] → arr[low+i]

low
high
..... 7 ......

low = 2   high = 2
mid: 2

(2, 2)

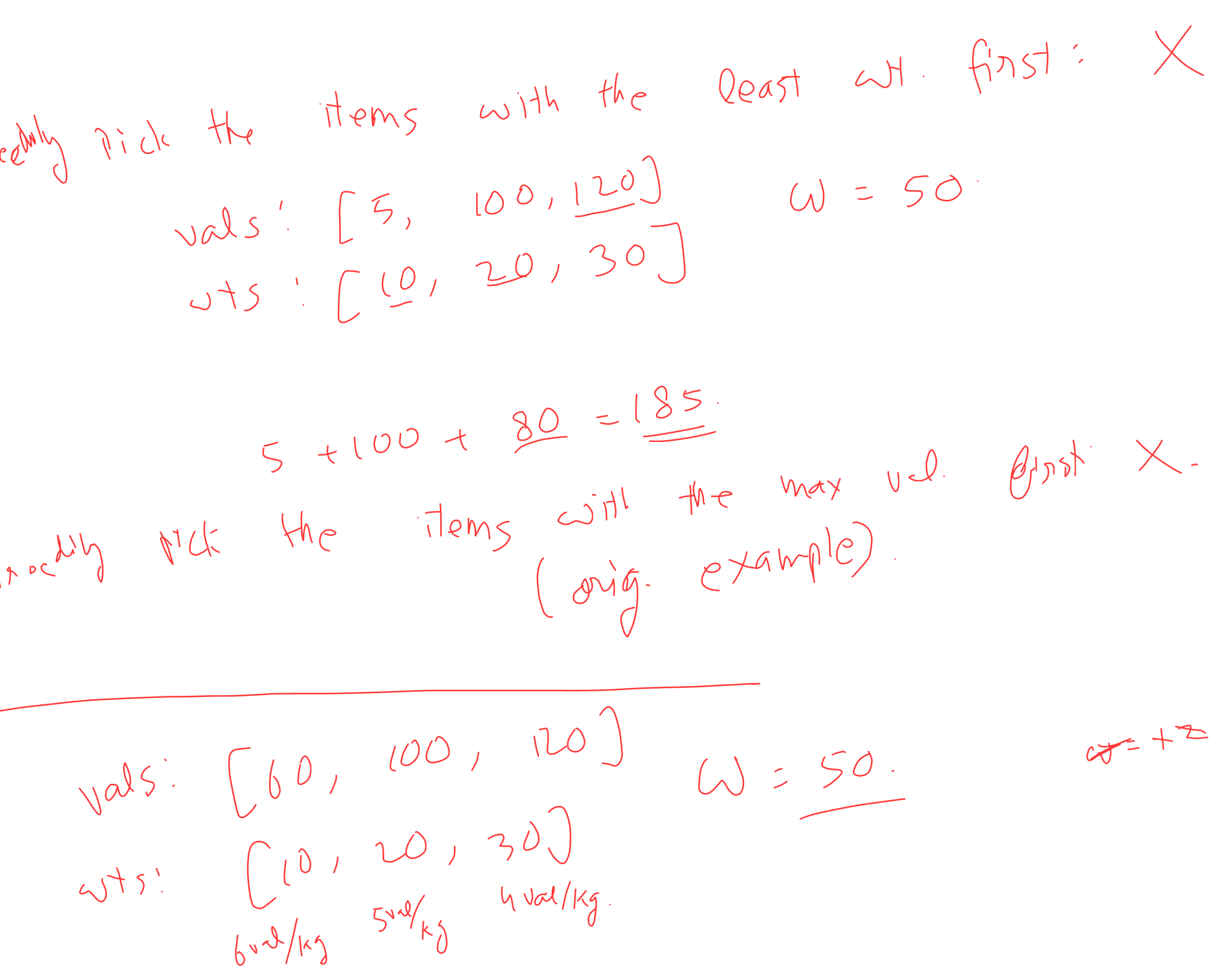
low = 1   high = 2
mid = ~~2~~ 1

0 1 2 3
[3, 5, 8, 2]

[3, 5]   [8, 2]   (8) (2)
[3]   [5]   [8]   [2]

low = 2   high = 3
mid = 2

4 temp, [2, 8]



T(n) = 2T(n/2) + n
↳ TC: O(n log n).

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vals: [60, 100, 120]
wts: [10, 20, 30]
W = 50 → 40 → 20 → 0

60 + 100 + 80 = 240

Greedily pick the items with the least wt first: X

vals: [5, 100, 120]
wts: [10, 20, 30]
W = 50

5 + 100 + 80 = 185

Greedily pick the items with the max val first X.
(orig. example)

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vals: [60, 100, 120]
wts: [10, 20, 30]
6val/kg 5val/kg 4val/kg

W = 50.

6/1 6/2 6/3 6/4 ... 6/10 5/11 5 ... 5/30 4 4 4 4 ... /50

60 + 100 + 80 = 240

{val/wt, wt}

~~[{6, 10}, {5, 20}, {4, 30}]~~
[{4, 30}, {5, 20}, {6, 10}]

W = ~~50~~ ~~40~~ ~~20~~ 0
res = 60 + 100 + 80

vals[i] → int
wts[i] → int
} → int

5/2 = 2

int/int → int

vals[i] = 5
wts[i] = 2

valByWt = 2.5
2

int/int → int

double/int OR int/double → double

[0, 30]
[5, 10]
[15, 20]

(0, 30) R1   (5, 10) R2   (15, 20) R2

If a meet ends at 't' & another starts at 't', then both can be accommodated in the same room.

O/P = 2

[1, 18] R1
[18, 23] R1
[15, 29] R2
[4, 15] R2
[2, 11] R3
[5, 13] R4

O/P = 4

BF.   TC = $O(n^2)$.

(1, 18) → R1
(2, 11) → R2
(4, 15) → R3
(5, 13) → R4
(15, 29) →
(18, 23) →

R1 (1, 18) (18, 23)
R2 (15, 29) (4, 15)
R3 (2, 11)
R4 (5, 13)

s s s s e e e s e s e e
1 2 4 5 11 13 15 15 18 18 23 29
1 2 3 4 3 2 1 2 1 2 1 0

min. No. of meeting rooms = Maximum no. of concurrent meetings taking place at any given point time

(5, s)   (5, e)

[[1, 5], [5, 10]] → O/P = 1

s s e e
1 5 5 10
1 2 1 0   X

s e s e
1 5 5 10   ✓

TODO: Train Platforms Problem.